

# Negative Exponent Law

## Lesson Plan

### 1. Lesson Information

|        |                                  |              |              |
|--------|----------------------------------|--------------|--------------|
| Course | Grade 7 Accelerated<br>(1205050) | Unit         | 3            |
| Lesson | Lesson 4 — Negative Exponent Law | Date         | _____        |
| Length | 53 minutes (one period)          | Benchmark(s) | MA.8.NSO.1.3 |

### 2. Standards

**MA.8.NSO.1.3** Extend previous understanding of the laws of exponents to include integer exponents. Apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to integer exponents and rational-number bases, with procedural fluency.

**Must.** Integer exponents, negatives included. Apply the laws to evaluate numerical expressions and to generate equivalent ones, with procedural fluency.

**Must not.** No fractional exponents. Bases are nonzero wherever a negative exponent appears; rational bases wait for 3.05.

### 3. Mathematical Thinking and Reasoning Standards

*the evidence for each, from this period*

| MTR            | What it asks                                                                        | The evidence in this lesson                                                                                            |
|----------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| MA.K12.MTR.5.1 | Use patterns and structure to help understand and connect mathematical concepts.    | Notes I — the halving table; students fill $2^{-1}$ , $2^{-2}$ and $2^{-3}$ from the pattern before the rule is named. |
| MA.K12.MTR.4.1 | Engage in discussions that reflect on the mathematical thinking of self and others. | Board 5 — when the room splits between $1/49$ and $-49$ , the sixty-second re-vote before the reveal.                  |
| MA.K12.MTR.6.1 | Assess the reasonableness of solutions.                                             | Board 1 — a reciprocal of $8^3$ is a small positive number, so $-512$ cannot be right; students say why.               |

### 4. Learning Target and Essential Question

**Learning target.** I can rewrite a power with a negative exponent using a positive exponent, and evaluate it.

**Essential question.** How do you evaluate a numerical expression that has a negative exponent?

**Building on.** MA.7.NSO.1.1 (3.01–3.03) — the seven laws with whole-number exponents; Grade 6 — reciprocals and dividing by a fraction.

**Working toward.** 3.05 — negative exponents on rational bases; 3.06–3.07 — several laws at once; MA.912.NSO.1.1 — rational exponents in Algebra 1.

## 5. Sequence

*read from the deck; never typed*

| Segment                             | Min       | What the teacher does                                                                        | What students do                           |
|-------------------------------------|-----------|----------------------------------------------------------------------------------------------|--------------------------------------------|
| Settle in; post the learning target | 1         | Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet.  | Copy the target.                           |
| Warm-Up                             | 5         | Two minutes silent, then reveal. One sentence of reteach per question at most.               | Four retrieval questions, alone, no notes. |
| Notes                               | 11        | Teach from the front. Expand once per law; say the OFF line out loud.                        | Copy the notes; answer the checks aloud.   |
| Example 1                           | 3         | Work the example; name the wrong boards before they appear.                                  | Watch, then work the Your Turn alone.      |
| Your Turn                           | 3         | Circulate; reveal and name what a wrong board most likely did.                               | Work it alone; boards up on the cue.       |
| Example 2                           | 3         | Work the example; name the wrong boards before they appear.                                  | Watch, then work the Your Turn alone.      |
| Your Turn                           | 3         | Circulate; reveal and name what a wrong board most likely did.                               | Work it alone; boards up on the cue.       |
| Whiteboards                         | 13        | One question at a time. Boards down until the cue, all up together; scan the back row first. | Answer on a board; hold it up on three.    |
| Independent Set                     | 6         | Silent. Circulate and mark what you see; do not teach.                                       | Six questions, written, alone.             |
| IXL                                 | 5         | Assign the day's IXL skills.                                                                 | Start the skills; finish tonight.          |
| <b>Total</b>                        | <b>53</b> |                                                                                              |                                            |

## 6. Gradual Release

**I do.** The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the OFF line is said out loud rather than printed.

**We do.** Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

**You do.** The nine-question whiteboard round (13 minutes) and then the six-question independent set, silent and written.

## 7. Higher-Order Questions

| Question                                                                                                                        | DOK   |
|---------------------------------------------------------------------------------------------------------------------------------|-------|
| Why does an exponent that keeps going down past zero give a positive value? Point to the row in the table where you can see it. | DOK 2 |
| Ellen writes $12^{-2} = -144$ . What would have to be true about the expression for $-144$ to be the answer?                    | DOK 3 |
| Two students evaluate $(-2)^{-3}$ and get $1/8$ and $-1/8$ . Which sign came from the base and which from the exponent?         | DOK 3 |

## 8. Checks for Understanding

each board, and what to do about it

| # | What the board asks                                                                                                                                                         | Answer                     | If the board is wrong                                                                                                                                                                                                                                                                             |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | $8^{-3}$                                                                                                                                                                    | $\frac{1}{8^3}$            | $-8^3$ or $-512$ — negative exponent read as a negative value [B1G-M 8.NSO.1.3 task: $-b$ vs $b^{-1}$ ]; $1/8$ — the exponent was dropped                                                                                                                                                         |
| 2 | A sheet of paper 30 centimeters long is folded in half 7 times. Each fold makes the top layer half as big. What fraction of the whole sheet is the top layer after 7 folds? | $\frac{1}{128}$            | 128 — the number of layers, not the fraction asked for [the question names the thing wanted]; $1/14$ — multiplied base by exponent; $-128$ — negative exponent read as a negative value [B1G-M 8.NSO.1.3]; anything built from 30 — the unneeded figure was used                                  |
| 3 | $\frac{1}{10^{-2}}$                                                                                                                                                         | $10^2 = 100$               | $1/100$ or $0.01$ — the negative exponent was made positive without moving the power [law applied in only one direction]                                                                                                                                                                          |
| 4 | $(-3)^{-3}$                                                                                                                                                                 | $-\frac{1}{27}$            | $1/27$ — the base's own sign was dropped [8.NSO.1.3: rational bases include negatives]; $-27$ — the negative exponent was treated as positive                                                                                                                                                     |
| 5 | $7^{-2}$ Which is the value?                                                                                                                                                | $B — 1/49$                 | (A) reciprocal of $7 \cdot 2$ instead of $7^2$ [reciprocal right, power wrong]; (C) the negative exponent was read as a negative value, $-(7^2)$ [B1G-M 8.NSO.1.3 task: $-b$ vs $b^{-1}$ ]; (D) multiplied the base by the exponent, $7 \cdot (-2)$ [misconception: exponent treated as a factor] |
| 6 | $4^{-3} \cdot 4^7$ Which expression is equivalent?                                                                                                                          | $D — 4^4$                  | (A) multiplied the exponents [confused with power of a power]; (B) multiplied the bases as well [7.NSO.1.1 misconception: operating on the bases]; (C) subtracted instead of adding, $-3 - 7$ [product law: add the exponents, including a negative one]                                          |
| 7 | $3^{-4} \cdot 3^4$                                                                                                                                                          | $1$                        | $0$ — zero exponent read as 0 [8.NSO.1.3: zero exponent law, $a^0 = 1$ ]; $81$ or $1/81$ — one factor was ignored                                                                                                                                                                                 |
| 8 | $\frac{11^{-4}}{11^2}$                                                                                                                                                      | $\frac{1}{11^6}$           | $11^{-2}$ or $1/121$ — subtracted $4 - 2$ ignoring the sign [integer subtraction, MA.7 prerequisite]; $11^6$ — the minus was dropped at the end                                                                                                                                                   |
| 9 | Ellen says the value of $12^{-2}$ is $-144$ . Explain why she is wrong, and show the correct value.                                                                         | $12^{-2} = 1/12^2 = 1/144$ | ' $-1/144$ ' — took the reciprocal AND the opposite [B1G-M 8.NSO.1.3 task: $-b$ vs $b^{-1}$ ]; 'she forgot parentheses' — confused $12^{-2}$ with $(-12)^2$                                                                                                                                       |

**MTR.4.1 — the one peer move.** When a board splits — no option holding about two-thirds of the room — sixty seconds of “convince the person next to you”, then re-vote, then reveal. The minute comes out of the whiteboard remainder.

## 9. Differentiation

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**ESE / IEP.** Hand the Notes I table already half filled —  $2^4$  through  $2^0$  complete — so the student's work is the last three cells and the pattern is visible, not remembered. The completed table stays on the desk for the whiteboard round.

**ELL.** The lesson turns on two English words. Put RECIPROCAL (flip it) and OPPOSITE (change the sign) side by side on the board with 'the reciprocal of 8 is  $1/8$ ' under one and 'the opposite of 8 is  $-8$ ' under the other, and point at them on every board that asks for a value.

**Enrichment.** Ask for  $(2/3)^{-2}$  a day before 3.05 introduces it, and for the  $n$  that makes  $2^n = 1/1024$ .

## 10. Closure, Vocabulary, Materials, Practice

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**Closure.** Board 9 is written work and it is the exit evidence: the student has to use the word reciprocal. Read the boards, not the papers.

**Vocabulary.** **negative exponent law; reciprocal; opposite**

**Materials.** Whiteboards and markers. Calculators are allowed on everything; on a calculator  $5^{-2}$  returns 0.04, which is  $1/25$  and worth showing once.

**Practice.** The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due at the start of the next class: Understanding negative exponents — YBB; Evaluate powers with negative exponents — WGS.